What a determinant measures

For square $A \in \mathbb{R}^{n \times n}$, $\det A$ gives a signed volume multiplier and tests invertibility.

$$A \text{ invertible} \iff \det A \neq 0.$$

2×2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

has signed area $ad - bc$. Independent columns give nonzero area with a sign determined by orientation. Collinear columns give zero area.

In n dimensions, $\det A$ is signed volume. Its absolute value gives the volume ratio and its sign records orientation reversal. The same quantity appears in inverse, pivot, and cofactor formulas.

A zero determinant tests exact singularity. Determinant magnitude alone does not measure conditioning. Chapter 11 treats scale and numerical rank.

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Three defining properties

Regard the determinant as a function of its columns

$$D(a_1, \ldots, a_n).$$

with these properties.

1. Normalization:

$$D(e_1, \ldots, e_n) = 1.$$

2. Alternation. Exchanging two columns reverses the sign.

$$D(\ldots, a_j, \ldots, a_k, \ldots) = -D(\ldots, a_k, \ldots, a_j, \ldots).$$

3. Multilinearity. The function is linear in each column separately.

$$D(\ldots, \alpha a_j + \beta b_j, \ldots) = \alpha D(\ldots, a_j, \ldots) + \beta D(\ldots, b_j, \ldots).$$

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Three defining properties (continued)

These properties determine the determinant uniquely. Expand each column in the standard basis and apply multilinearity. Terms with repeated basis vectors vanish. The surviving terms are permutations of the standard basis, whose values are their permutation signs. This gives the permutation formula below, which also satisfies all three properties, proving existence and uniqueness. This chapter uses $\mathbb{R}$ or $\mathbb{C}$. In characteristic 2, alternation must explicitly require zero for repeated columns, rather than relying only on sign reversal.

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Consequences for row and column operations

When two columns coincide, exchanging them leaves the matrix unchanged but reverses the determinant.

$$D = -D,$$

Thus $D = 0$ over a field of characteristic other than 2.

Adding a multiple of one column to another gives

$$D(\ldots, a_j + ca_k, \ldots) = D(\ldots, a_j, \ldots) + cD(\ldots, a_k, \ldots).$$

The second term has repeated columns and vanishes. Column addition preserves the determinant. Transpose invariance will give the corresponding row rule.

Dependent columns therefore give zero determinant. Express one as a combination of the others; multilinearity expands the determinant into terms with repeated columns.

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Scaling, exchanges, and triangular matrices

Scaling one column by λ gives

$$\det(\ldots, \lambda a_j, \ldots) = \lambda \det(\ldots, a_j, \ldots).$$

Consequently,

$$\det(cA) = c^n \det A.$$

This agrees with scaling every edge of an n-dimensional body by c.

A row exchange multiplies the determinant by -1. For upper triangular U,

$$\det U = u_{11} u_{22} \cdots u_{nn}.$$

A zero diagonal entry of a triangular matrix makes its determinant zero and the matrix singular. This connects determinants to elimination.

Track scaling and exchanges in an implementation to preserve the determinant's value and sign.

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$\det A = \det A^T$

Permutation formula：

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

After transposing, set $\tau = \sigma^{-1}$.

$$a_{\sigma(i),i}$$

becomes $a_{j,\tau(j)}$, and

$$\operatorname{sgn}(\tau) = \operatorname{sgn}(\sigma).$$

therefore

$$\det A^T = \det A.$$

Rows and columns obey the same determinant rules. The following nonnegativity claim assumes $A \in \mathbb{R}^{n \times n}$.

$$\det(A^T A) = (\det A)^2 \geq 0.$$

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$\det A = \det A^T$ (continued)

For every real $A \in \mathbb{R}^{m \times n}$, $A^T A$ is positive semidefinite, and positive definite exactly when $\operatorname{rank} A = n$. For an invertible square matrix, both conditions hold. For complex matrices, transpose does not conjugate, and squares need not be nonnegative. For $A = [i]$, $A^T A = [-1]$, giving $z^T A^T A z = -1$ at $z = 1$.

For complex matrices, use the conjugate transpose $A^* = \overline{A}^T$.

$$\det(A^* A) = \overline{\det A} \det A = |\det A|^2 \geq 0, \qquad z^* A^* A z = \|Az\|_2^2 \geq 0.$$

Then $A^* A$ is Hermitian positive semidefinite and is positive definite exactly at full column rank. The energy identity still holds for rectangular $A \in \mathbb{C}^{m \times n}$, but $\det A$ is undefined, so the square determinant product cannot be used.

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Real Gram matrices

Real Gram: A^TA is symmetric positive semidefinite

$A \in \mathbb{R}^{m \times n}$ ，$z^TA^TAz = \|Az\|_2^2 \geq 0$

A^TA is always symmetric PSD

rank(A)=n ⇔ A^TA is positive definite

$\det(A^TA) = \det(A)^2$ requires square A.

A=[[1,1],[1,2]]

det A=1

$\det(A^TA) = 1^2 = 1$

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Complex Gram matrices

Complex Gram: transpose versus conjugate transpose

$A=[i]$

$A^TA=[-1]$ · $\det=-1$

$z=1$: $z^TA^TAz=-1$

Transpose Gram need not be PSD

$A^*=[-i]$ · $A^*A=[1]$

$\det(A^*A)=|\det A|^2=1$

$z^*A^*Az=\|Az\|_2^2\geq 0$

A^*A is Hermitian PSD

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Products and inverses

Fix B and consider

$$F(A) = \det(AB)$$

which is multilinear and alternating in the rows of A, with

$$F(I) = \det B.$$

Every alternating multilinear function is its value at the identity times the determinant, even when that value is zero. Therefore

$$\det(AB) = \det A \det B.$$

For invertible A,

$$1 = \det(A^{-1}A) = \det(A^{-1}) \det A,$$

so

$$\det(A^{-1}) = 1/\det A.$$

If either matrix is singular, the product has deficient rank and both determinant expressions vanish.

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Products and inverses (continued)

In general, $\det(A+B) \neq \det A + \det B$. Multilinearity in individual rows or columns is not additivity on the matrix space.

For example, let $A = I_2$, $B = \operatorname{diag}(2,3)$. Then $\det A = 1$, $\det B = 6$, but

$$\det(A+B) = \det \operatorname{diag}(3,4) = 12 \neq 1 + 6.$$

Multilinearity splits one row or column while the others stay fixed. It does not split a sum of entire matrices.

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Elimination and pivots

If elimination reaches upper-triangular U using only row additions, without row exchanges or scalings,

$$\det A = \det U = u_{11} u_{22} \cdots u_{nn}.$$

With q row exchanges,

$$\det(PA) = (-1)^q \det A,$$

compute the pivot product of PA and multiply by $(-1)^q$.

For example,

$$A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}.$$

Swap rows：

$$PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}, \qquad \det(PA) = 6.$$

Thus $\det A = -6$, agreeing with $ad - bc = 0 \cdot 4 - 2 \cdot 3 = -6$.

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Elimination and pivots (continued)

Rank often tests invertibility more directly; signed-volume calculations require the exchange sign.

For completeness, dependent columns make one column a combination of the others. Column multilinearity then expands the determinant into terms with repeated columns, all zero. Therefore $\det A \neq 0$ forces independent columns, a basis of $\mathbb{F}^n$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and invertibility. Conversely, invertibility gives

$$\det A \det(A^{-1}) = \det(AA^{-1}) = 1.$$

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Elimination: two conditions to track

Account for row scaling. For $A = \operatorname{diag}(2, 3)$, $\det A = 6$. Dividing row 1 by 2 gives $U = \operatorname{diag}(1, 3)$, with $\det U = 3$; multiply by 2 to recover the original determinant.

A zero candidate pivot does not itself mean singularity. For $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the first candidate is zero. A row swap produces I, and the swap gives $\det B = -1$. Singularity follows only if an adequate nonzero pivot cannot be found and rank is deficient.

In general $PA = LU$ gives $\det A = \det(P)^{-1} \det L \det U$. The factor $\det L$ can be omitted only when L is unit lower triangular; each row swap reverses the sign of $\det P$.

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The permutation formula

The inversion count $N(\sigma)$ counts pairs with $i < j$ and $\sigma(i) > \sigma(j)$. Define

$$\operatorname{sgn}(\sigma) = (-1)^{N(\sigma)}.$$

Big formula:

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)}.$$

Each term selects one entry from every row and every column. In size 3×3,

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

The first three permutations are even and the last three are odd. The Sarrus pattern is specific to dimension 3 and does not extend unchanged to larger matrices.

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Cofactor expansion

Delete row i and column j to form minor matrix M_{ij}. Its signed determinant is the cofactor

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

Expansion along row i gives

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

An analogous formula holds along any column.

For example,

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix}.$$

Expanding along row 1 gives

$$\det A = 2(12 - 2) - 1(-4) + 0 = 24.$$

Expanding along row 2 gives

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Cofactor expansion (continued)

$$(-1)(-4) + 3(8) + 1(-4) = 24.$$

Agreement follows from the cofactor identity, rather than from a special property of this example.

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Deriving cofactor expansion

Expand row i as

$$r_i = \sum_{j=1}^{n} a_{ij} e_j^T.$$

Row multilinearity gives

$$\det A = \sum_{j=1}^{n} a_{ij} D(A_{i \leftarrow e_j}).$$

In $A_{i \leftarrow e_j}$, row i has a single 1 in column j. Move that row to the top with $i - 1$ adjacent swaps and that column to the left with $j - 1$ adjacent swaps. The first row is now $(1, 0, \ldots, 0)$. Clear the rest of the first column by row additions; the remaining lower-right block is M_{ij}.

Consequently,

$$D(A_{i \leftarrow e_j}) = (-1)^{i+j} \det M_{ij} = C_{ij}.$$

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Deriving cofactor expansion (continued)

Summing the terms gives cofactor expansion. The checkerboard signs record the parity of those adjacent swaps.

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Pivots and a tridiagonal recurrence

If pivoted elimination gives $PA = LU$ with unit lower-triangular L,

$$\det(PA) = \prod_i u_{ii},$$

$$\det A = (-1)^q \prod_i u_{ii}.$$

A zero leading principal minor need not mean singularity; a row exchange may suffice.

For the tridiagonal matrix,

$$T_n = \begin{bmatrix} 2 & -1 & 0 & \cdots \\ -1 & 2 & -1 & \ddots \\ 0 & -1 & 2 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{bmatrix}.$$

expand along the first row.

$$D_n = 2D_{n-1} - D_{n-2}, \qquad n \geq 2.$$

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Pivots and a tridiagonal recurrence (continued)

With initial values $D_0 = 1, D_1 = 2$, where $D_0 = 1$ is the empty-determinant convention, induction gives $D_n = 2n - (n-1) = n + 1$:

$$D_n = n + 1.$$

All leading principal minors are positive, also proving that T_n is positive definite.

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Block elimination: the Schur complement

Let $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, where A is invertible and square, D is square, and all block sizes are compatible. Block elimination gives

$$L = \begin{bmatrix} I & 0 \\ -CA^{-1} & I \end{bmatrix}, \qquad LM = \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix}.$$

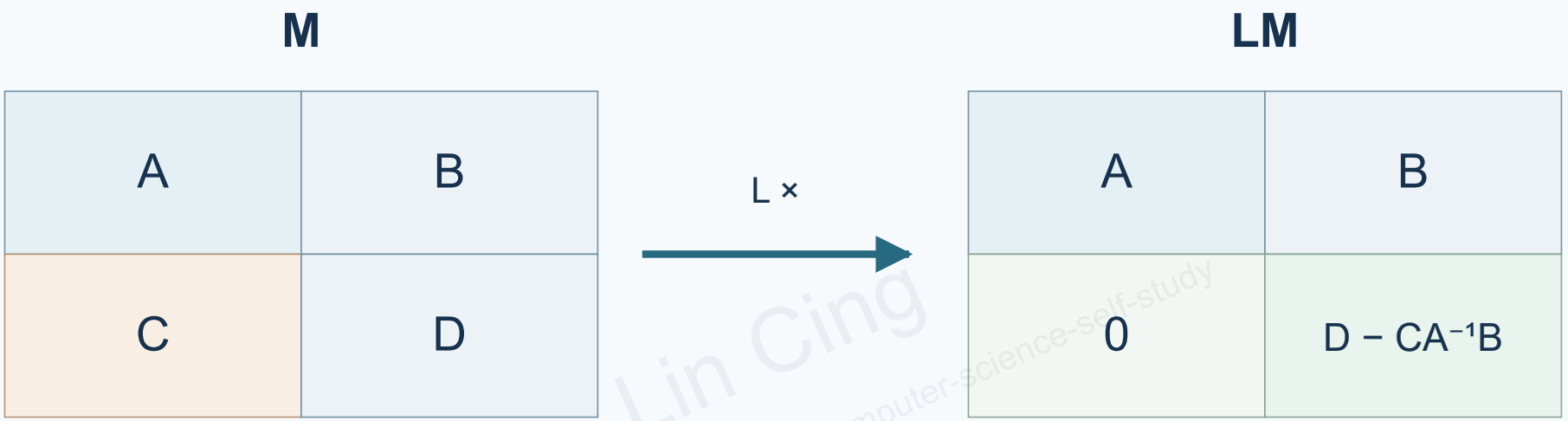
Here $\det L = 1$, and a block upper-triangular determinant is the product of the determinants of its diagonal blocks. Thus

$$\det M = \det A \, \det(D - CA^{-1}B).$$

The second factor is the Schur complement of A. Matrix factors cannot be reordered freely; when A is singular, this formula's A^{-1} is undefined.

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The block elimination step



The lower-left block becomes $-CA^{-1}A + C = 0$; the lower-right simultaneously becomes $D - CA^{-1}B$. The identity diagonal blocks give $\det L = 1$.

Why the scalar formula $AD - BC$ can fail

Take $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Here $A^{-1} = A$, $\det A = -1$, and

$$D - CA^{-1}B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \det M = (-1)(-2) = 2.$$

But $AD - BC = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$ has determinant 1. Reordering noncommuting blocks without justification changes the answer.

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The adjugate

Cofactor matrix $C = [C_{ij}]$, adjugate:

$$\operatorname{adj} A = C^T.$$

Cofactor identities:

$$\sum_{j=1}^{n} a_{ij} C_{kj} = \begin{cases} \det A, & i = k, \\ 0, & i \neq k. \end{cases}$$

For $i \neq k$, the sum is the determinant after replacing row k by row i. The repeated rows make it zero.

Hence

$$A(\operatorname{adj} A) = (\det A) I.$$

If $\det A \neq 0$,

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A.$$

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The adjugate (continued)

2×2:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

An adjugate requires n^2 minors, so it is mainly useful for identities and small matrices.

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Cramer's rule

Suppose A is invertible and $Ax = b$. Replace column j of the identity by x. Linearity and repeated-column cancellation give

$$x_j = \det(e_1, \dots, e_{j-1}, x, e_{j+1}, \dots, e_n).$$

Multiply this matrix by A and use the determinant product rule to obtain

$$(\det A)x_j = \det(a_1, \dots, a_{j-1}, b, a_{j+1}, \dots, a_n).$$

Thus

$$x_j = \frac{\det A_j}{\det A},$$

where A_j replaces column j of A by b.

For example,

$$\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Here $\det A = -2$, $\det A_1 = -4$, and $\det A_2 = 2$, giving

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Cramer's rule (continued)

$$x_1 = 2, \qquad x_2 = -1.$$

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Limits of Cramer's rule

Cramer's rule makes each x_j linear in b, since the determinant is linear in the replaced column.

Computing n determinants separately is more costly than one pivoted LU factorization for many numerical solves. A small nonzero denominator alone gives no reliability guarantee.

When $\det A = 0$, Cramer's quotient is undefined. Use ranks and the augmented matrix to distinguish

- no solution;
- infinitely many solutions.

A zero determinant states singularity of the square coefficient matrix. It does not state incompatibility of a particular right-hand side.

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Determinants as signed areas

Let

$$u = (a, c)^T, \qquad v = (b, d)^T,$$
$$P = [u\ v] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Parallelogram area：

$$|\det P| = |ad - bc|.$$

Without the absolute value, the sign records the orientation from u to v.

For

$$u = (2, 1)^T, \qquad v = (1, 3)^T,$$

Then

$$\det P = 6 - 1 = 5,$$

both the signed area and the area are 5.

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Determinants as signed areas (continued)

For the linear map $x \mapsto Px$,

$$\operatorname{area}(P(S)) = |\det P| \operatorname{area}(S).$$

if $\det P = 0$, the image has dimension at most one and area zero.

The diagram maps the unit-square edges $(1,0)^T, (0,1)^T$ to $u = (2,1)^T, v = (1,3)^T$. Their respective areas are

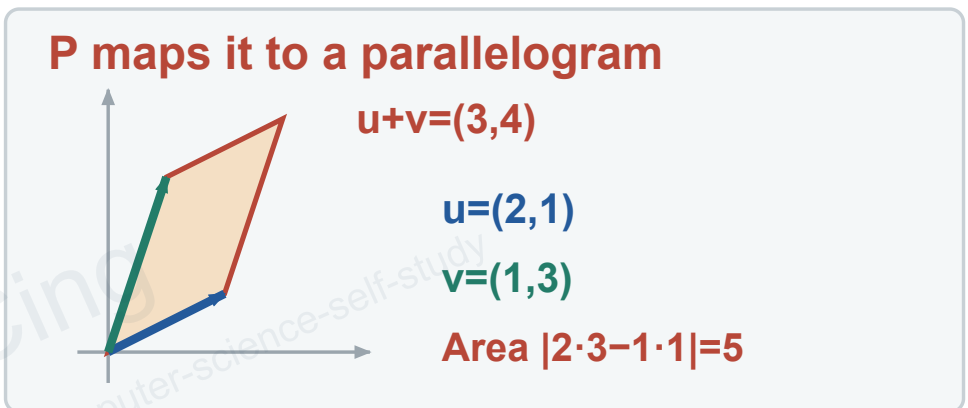
$$|\det I| = 1, \qquad |\det P| = |2 \cdot 3 - 1 \cdot 1| = 5.$$

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Signed area and linear maps

Signed area: det([[2,1],[1,3]])=5





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Volume and orientation in n dimensions

For parallelepiped edges $a_1, \ldots, a_n$ placed as columns,

$$\text{volume} = |\det A|.$$

This uses standard coordinate units. Convert incompatible units before interpreting physical volumes.

Exchanging edges reverses signed orientation but preserves physical volume. Adding a multiple of one edge to another is a shear and preserves both.

Elimination tracks volume through row scaling and exchanges, while the pivot product supplies the resulting signed determinant.

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Cross products and determinants

In $\mathbb{R}^3$,

$$w = u \times v$$

is perpendicular to u, v, with norm equal to their parallelogram area. Its components are

$$w_1 = u_2 v_3 - u_3 v_2,$$

$$w_2 = u_3 v_1 - u_1 v_3,$$

$$w_3 = u_1 v_2 - u_2 v_1.$$

Hence

$$w^T u = 0, \qquad w^T v = 0.$$

Antisymmetry gives

$$v \times u = -u \times v$$

the same sign change as exchanging determinant columns.

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Cross products and determinants (continued)

The formal determinant using i, j, k is a mnemonic for cofactor directions. Basis vectors are not scalar entries in the same field. The component formulas provide an unambiguous definition.

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A numerical example

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix}.$$

Cofactor expansion along row 1 gives

$$\det A = 1(1-3) - 2(0-6) = -2 + 12 = 10.$$

Adding twice column 1 to column 2 produces B with the same determinant, 10. The operation is a shear.

The columns span a parallelepiped with signed volume 10. Exchanging two columns reverses the sign. Dependence collapses the volume to zero.

Adjugate and Cramer computations can check small examples. Computing n^2 minors is unsuitable for a large numerical inverse.

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Chapter review

- Multilinearity, alternation, and normalization determine the determinant.
- Track swaps and scaling during elimination; check the diagonal of L in $PA = LU$.
- Real Gram matrices use transpose; complex Gram matrices use conjugate transpose.
- A nonzero determinant means invertibility, but its magnitude alone does not measure conditioning.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 5.

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